

RESEARCH BRIEFING

Are You Getting What You Pay For?

Auditing Model Substitution in LLM APIs:
Why Software Tests Fail and Hardware Attestation Wins

Based on: arxiv.org/abs/2504.04715

Audience: ML Researchers · AI Security Practitioners · Cloud Infrastructure Engineers

Commercial API providers may covertly serve cheaper models —
and current detection methods cannot reliably catch them.

The Model Substitution Problem: Economic Incentives Drive Covert Swaps

Formal hypothesis-testing framework | Four substitution types

Why Substitution Happens

- Providers charge for named models (e.g. Llama-3-70B, Qwen2-72B)
- APIs are black boxes — users cannot inspect execution
- Quantized / smaller models cut GPU costs significantly
- Pricing stays identical; revenue impact is zero
- No current mechanism enforces model identity

$H_0: P_{\text{actual}}(y|x) = P_{\text{spec}}(y|x)$ vs $H_1: P_{\text{actual}} \neq P_{\text{spec}}$

Four Substitution Types

1 Smaller Model

e.g. 8B served instead of 70B · ~8× cheaper to run

2 Quantized Variant

FP8/INT8 instead of FP16 · Lower precision, similar scores

3 Fine-tuned Version

Updated or domain-specific weights · Behaviour shifts undetected

4 Different Family

Entirely different architecture · No contractual equivalence

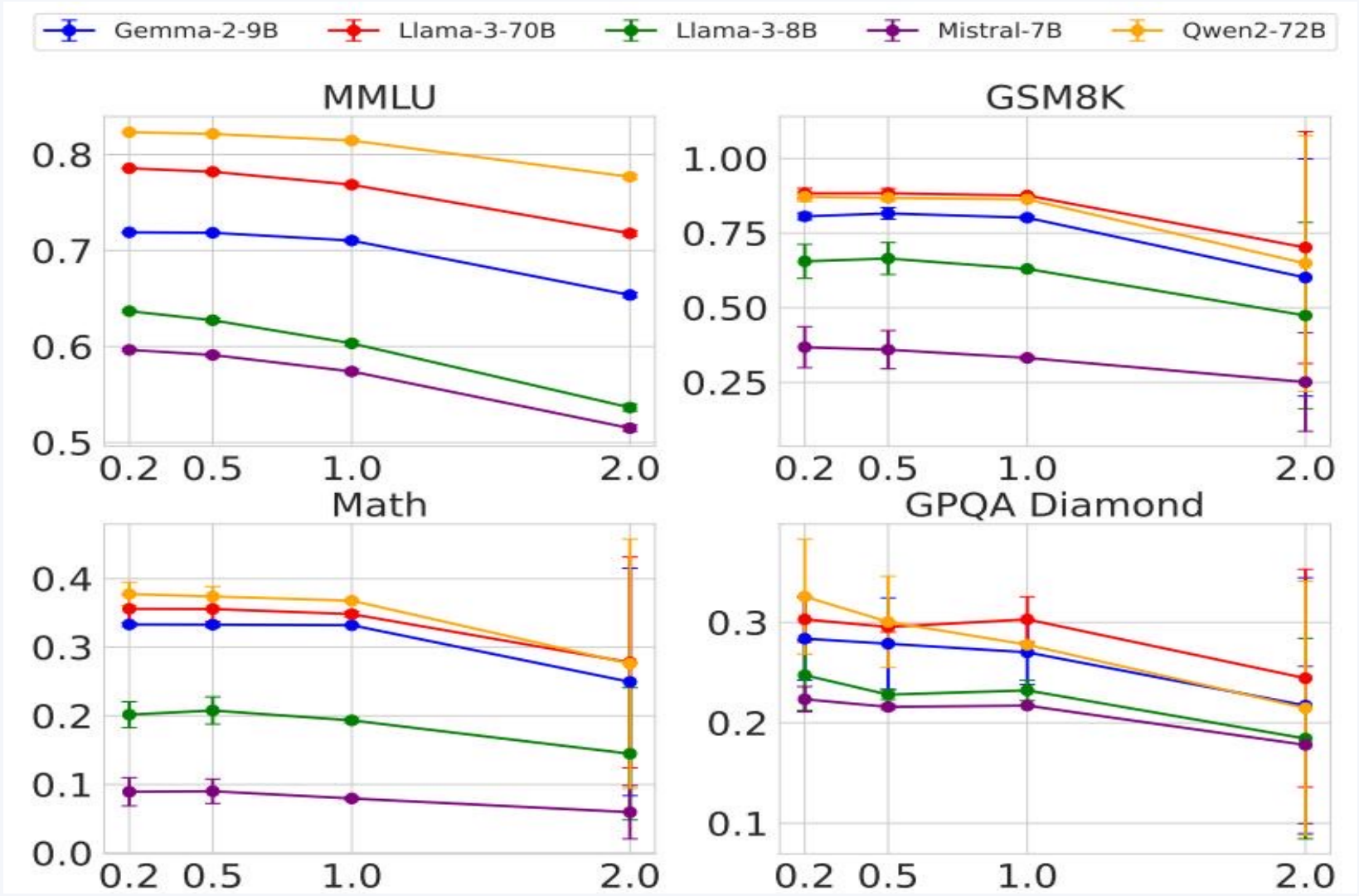
FP8 Quantization Preserves Most Accuracy — Making Substitution Hard to Detect

Original vs FP8 across MMLU · GSM8K · MATH · GPQA Diamond

Model	MMLU	GSM8K	MATH	GPQA Diamond
Llama-3-8B-Instruct	62.69 ± 0.18	61.14 ± 3.47	20.65 ± 3.43	22.62 ± 0.22
Llama-3-8B-Instruct-FP8	62.43 ± 0.26	60.90 ± 4.10	14.91 ± 2.49	20.14 ± 0.24
Llama-3-70B-Instruct	78.05 ± 0.08	88.06 ± 1.44	35.69 ± 1.33	29.60 ± 0.51
Llama-3-70B-Instruct-FP8	77.88 ± 0.13	87.35 ± 1.24	35.75 ± 1.16	33.12 ± 0.30
Gemma-2-9b-it	71.86 ± 0.08	81.80 ± 1.35	33.41 ± 0.28	28.64 ± 2.97
Gemma-2-9b-it-FP8	71.92 ± 0.11	79.41 ± 1.14	32.53 ± 0.34	27.81 ± 3.22
Qwen2-72B-Instruct	82.18 ± 0.08	86.72 ± 1.00	37.39 ± 1.41	29.93 ± 2.71
Qwen2-72B-Instruct-FP8	81.98 ± 0.08	86.82 ± 0.97	37.67 ± 1.39	31.08 ± 1.96
Mistral-7B-Instruct-v0.3	59.15 ± 0.10	35.90 ± 4.54	8.94 ± 1.24	21.60 ± 0.17
Mistral-7B-Instruct-v0.3-FP8	58.77 ± 0.13	32.20 ± 4.02	7.68 ± 1.23	22.72 ± 0.19

Higher Temperature Amplifies Performance Gaps but Increases Noise

Benchmark accuracy vs temperature | All five model families | figure_1



Key Observations

Temp 0.2–0.5

Tightest separation between original and FP8 — but still narrow

Temp 1.0–2.0

Performance degrades across all models; signal masked by noise

Detection gap

Higher temps don't reliably widen the original–FP8 gap; variance dominates

Implication

No temperature setting reliably enables statistical separation

↑ Temperature increases output variance, not detection power.

Log Probability Fingerprinting Fails Due to Inference Nondeterminism

Same model · Different framework / hardware → different log-probs | figure_5

Model	BERT Acc	T5 Acc	GPT2 Acc	LLM2Vec Acc
Llama3-70B-Instruct-FP8	50.55	50.10	50.45	49.90
Llama3-70B-Instruct-INT8	51.60	49.90	51.30	50.25
Gemma2-9b-it-FP8	49.95	50.30	51.20	49.80
Gemma2-9b-it-INT8	49.00	49.70	51.65	49.55
Mistral-7b-v3-Instruct-FP8	50.55	49.75	48.70	48.75
Mistral-7b-v3-Instruct-INT8	49.70	50.75	50.50	51.15
Qwen2-72B-Instruct-FP8	50.05	50.25	48.75	49.80
Qwen2-72B-Instruct-INT8	50.75	50.55	49.45	50.20

Why This Defeats Fingerprinting

Software variation

transformers 4.40 vs 4.50
vLLM 0.6.2 vs 0.8.2
→ Different float rounding per token

Hardware variation

NVIDIA A100 vs H100
→ Different CUDA kernel paths

Auditor's dilemma

Cannot distinguish legitimate infra change from covert model swap

Result

Log-prob distance between same model on different infra ≈ distance to a substitute

Production nondeterminism is indistinguishable from model substitution.

Randomized Routing at Low Rates Makes Statistical Detection Query-Prohibitive

Adversarial countermeasure: mixing p -fraction of substitute responses

The Randomized Substitution Attack

Provider routes a fraction p of queries to a cheaper substitute model, while serving the genuine model for the remaining $(1-p)$ queries.

Mixed distribution:

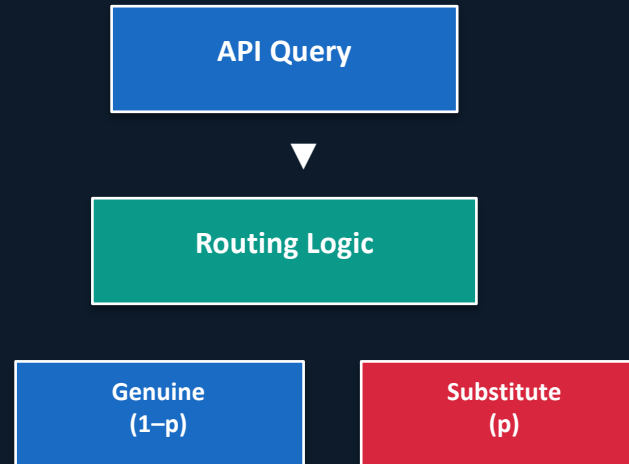
$$P_{\text{mix}}(y|x) = (1-p) \cdot P_{\text{spec}} + p \cdot P_{\text{sub}}$$

As $p \rightarrow 0$:

$$P_{\text{mix}} \rightarrow P_{\text{spec}}$$

Statistical distinguishability vanishes. The auditor must send exponentially more queries to achieve the same detection power as p decreases.

Query Routing Illustration



Benchmark Evasion

Known benchmark prompts detected → routed to genuine model
Regular traffic silently served by substitute

↑ p decreases $\Rightarrow$ detection requires exponentially more queries — practically infeasible

Software-Only Auditing Methods Are Fundamentally Unreliable

Three failure modes · Comparison matrix

Method Type	Core Approach	Weakness	Practical Limitation
Text-Based Statistical Tests	MMD and similar tests on text output distributions	Require enormous query budgets; fail against subtle substitutions	Economically prohibitive; easily defeated by small p
Log-Probability Fingerprinting	Compare token log-probs with local reference model	Production nondeterminism (SW versions, GPU hardware) swamps signal	Cannot distinguish infra change from actual model swap
Adversarial Countermeasures	Benchmark evasion + randomized routing	Provider actively routes known probes to genuine model	All software methods become query-prohibitive or blind
Verdict: Software-only approaches provide no provable guarantee of model identity under adversarial conditions.			

Trusted Execution Environments Provide Provable Cryptographic Model Integrity

Hardware attestation · NVIDIA Confidential Computing · Modest overhead

How TEEs Work

- **Hardware-secured enclave**
Isolated memory region; even the OS/hypervisor cannot read it
- **Cryptographic attestation**
Hardware signs a report: these weights + this code are loaded
- **Remote verification**
Third-party auditor verifies the signature without seeing private data
- **NVIDIA H100 support**
Confidential computing available today via NVIDIA CC mode
- **Modest overhead**
Small performance cost; no ZKP complexity or key-management burden

TEE vs Software Methods

Property	Software Only	TEE Attestation
Proof type	Statistical	Cryptographic
Adversarial robust?	No	Yes
Query budget	Exponential	Zero (one-time setup)
Nondeterminism safe?	No	Yes
Deployable today?	Yes (but useless)	Yes (NVIDIA CC)
Overhead	High query cost	Modest compute

TEEs are the only currently viable path to verifiable LLM API integrity.

Key Takeaways: Close the Verification Gap with Hardware Attestation

1

Model substitution is economically rational and practically undetectable via software

FP8 quantization cuts GPU costs significantly while benchmark scores barely change — providers have clear incentive.

2

FP8 quantization barely affects benchmarks — making it the perfect covert substitute

Scores fall within error bars for most model families on MMLU and GSM8K; MATH shows slightly larger gaps.

3

Inference nondeterminism defeats log-probability fingerprinting

Same model on different SW versions (transformers, vLLM) or GPU (A100/H100) produces divergent log-probs.

4

Randomized routing and benchmark evasion make all statistical tests query-prohibitive

As substitution rate $p \rightarrow 0$, detection requires exponentially more queries; known probes can be routed away.

5

TEEs are the only currently viable path to verifiable LLM inference

Hardware attestation gives cryptographic proof of model identity with modest overhead — deployable today on NVIDIA H100.

Call to Action: Demand TEE-backed attestation from LLM API providers. Standardise cryptographic model identity verification as a cloud service requirement.